PROJECT PHASE 1 – MAJOR PROJECT PROPOSAL

Title of the Project

IoT-Based Smart Waste Monitoring and Waste Level Prediction System Using Machine Learning.

1. Introduction

Waste management has become a major challenge in urban and semi-urban areas because conventional garbage collection systems follow fixed schedules instead of monitoring the actual condition of the bins. As a result, some bins overflow before collection, while others are emptied unnecessarily. Overflowing bins create unhygienic conditions, bad odour, environmental pollution, and health risks.

To solve this problem, this project proposes an IoT-based smart waste monitoring system that continuously measures the waste level inside garbage bins using sensors. The collected sensor data is sent to the cloud using an IoT module, and machine learning is used to predict when the bin will become full.

Unlike existing systems that only notify when the bin is already full, this project predicts the filling time in advance. This helps waste collection authorities plan efficient collection schedules and avoid overflowing bins.

2. Problem Statement

Traditional waste collection systems have the following limitations:

- Garbage bins are checked manually.

- Waste collection follows fixed timing, not actual waste level.

- Overflowing bins lead to bad smell and pollution.

- Existing smart bins only provide notification after the bin becomes full.

- Most previous systems do not predict future waste level accurately.

- Existing prediction systems use simple average methods and have high prediction error.

Therefore, there is a need for an intelligent waste monitoring system that not only detects the current level of garbage but also predicts future filling using machine learning.

3. Proposed Solution

The proposed system uses:

- Ultrasonic sensor to detect the waste level inside the bin.

- ESP32 microcontroller to process and send data.

- RTC module to record time and date.

- IoT platform for real-time monitoring.

- Hybrid Machine Learning algorithm for waste level prediction.

The system continuously monitors the garbage level and sends the data to a cloud platform. Based on previous sensor readings and time logs, the machine learning model predicts when the garbage bin will become full.

When the predicted waste level reaches a threshold, a notification is sent to the user or municipal worker.

4. Research Gap

After studying the reference papers, the following research gaps were identified:

Existing System	Limitation
IoT-based waste monitoring systems	Only detect the current waste level and cannot forecast future filling.
Systems using ultrasonic sensors and Blynk	Provide monitoring only; no accurate future waste prediction is available.
RTC-based smart waste systems	Use simple average-based prediction, resulting in higher prediction error.
Previous machine learning-based waste systems	Mainly focus on monitoring and alert generation, not future prediction.
Existing smart bins	Do not classify filling patterns based on day, time, or type of waste.

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Gap Identified

The existing systems monitor only the present condition of the bin. They do not intelligently predict the future waste level with better accuracy.

Proposed Improvement

This project introduces a machine learning model that uses:

- Waste level history

- Day of the week

- Time of day
- Recyclable fraction
- Organic / dry waste category
- Previous filling pattern

This improves prediction accuracy and helps in better scheduling of garbage collection.

5. Objectives

The main objectives of the project are:

1. To design an IoT-based smart waste monitoring system.
2. To measure waste level using ultrasonic sensors.
3. To store sensor readings with time and date.
4. To send real-time data to a cloud platform.
5. To develop a machine learning model for predicting future waste level.
6. To notify users when the bin is likely to become full.
7. To reduce overflowing bins and improve waste collection efficiency.

6. Scope of the Project

The scope of this project includes:

- Monitoring the garbage level in one or more smart bins.
- Recording time-based sensor data.
- Predicting future waste filling.

The project does not include:

- Automatic mechanical cleaning of the bin.
- Physical transportation of garbage.
- Large-scale municipal deployment.

7. Literature Survey

Paper 1

An Internet of Things Based Waste Management System Using Hybrid Machine Learning Technique

Findings

- Uses IoT sensors and machine learning.
- Predicts garbage level using historical data.
- Improves waste collection planning.

Limitation

- Prediction accuracy can still be improved.
- Less focus on real-time notification and time logging.

Paper 2

Trash Tracker: IoT Based Garbage Monitoring System Using Raspberry Pi

Findings

- Uses Raspberry Pi and ultrasonic sensor.
- Sends notification when bin becomes full.

Limitation

- Only detects full condition.
- No prediction model is used.

Paper 3

Development of an IoT-Based Smart Waste System with RTC Time Logging for Cleaning Assistance

Findings

- Uses RTC module for recording bin status.
- Gives notifications using IoT platform.
- Predicts filling time using average interval.

Limitation

- Prediction error is high.
- Uses only simple average instead of machine learning.

8. Methodology

The project methodology contains the following stages:

1. Sensor data collection

2. Data preprocessing

3. Feature selection

4. Machine learning model training

5. Prediction generation

6. IoT dashboard update

7. Notification sending

Step-by-Step Working :

Step 1: Data Collection

The ultrasonic sensor measures the distance between the top of the bin and the garbage.

Step 2: Data Transmission

ESP32 sends the measured values to cloud storage.

Step 3: Time Logging

RTC module records the exact date and time of each reading.

Step 4: Data Preparation

The collected data is converted into features such as:

- Day
- Time
- Waste category
- Recyclable fraction
- Previous waste level

Step 5: Machine Learning

The dataset is trained using different machine learning algorithms.

Possible algorithms:

- Random Forest
- Decision Tree
- Gradient Boosting
- XGBoost (optional)

Step 6: Best Model Selection

The model with highest accuracy and lowest RMSE is selected.

9. Hardware Requirements

Component	Purpose
ESP32	Main controller
HC-SR04 Ultrasonic Sensor	Measure garbage level
RTC DS3231 Module	Time logging
Breadboard	Circuit connection
Jumper Wires	Connecting components
Power Supply / Battery	Power source
Wi-Fi Connection	Send data to cloud

10. Software Requirements

Software	Purpose
Python	Data analysis and machine learning
HC-SR04 Ultrasonic Sensor	Model development
Arduino IDE	ESP32 programming
Blynk / ThingSpeak	IoT dashboard
VS Code	Coding
Pandas	Data preprocessing
Scikit-learn	Machine learning
NumPy	Numerical operations
Matplotlib	Graph plotting

11. Expected Outcome

The expected outputs of the project are:

- Real-time waste level monitoring.

- Accurate prediction of when the bin becomes full.

- Better garbage collection scheduling.

- Reduced manual checking and improved cleanliness.

12. Advantages of Proposed System

- Real-time monitoring.
- Intelligent future prediction.
- Reduces overflowing bins.
- Low-cost and easy to implement.
- Saves time and manpower.
- Better than fixed-time garbage collection systems.

13. Future Enhancement

Future improvements can include:

- Solar-powered smart bin.
- GPS-based route optimization for garbage vehicles.
- Camera-based waste classification.
- Deep learning for more accurate prediction.
- Mobile application for municipal workers.
- Deployment in smart cities.

14. Conclusion

The proposed project provides an effective solution for smart waste management using IoT and machine learning. The system not only monitors the current garbage level but also predicts future filling time. This helps in avoiding overflow, improving sanitation, and reducing unnecessary waste collection trips.

By introducing a hybrid machine learning model and time-based prediction, this project improves upon the limitations of existing systems and provides a better approach for smart city waste management.

15. References

1. An Internet of Things Based Waste Management System Using Hybrid Machine Learning Technique.

2. Trash Tracker: IoT Based Garbage Monitoring System Using Raspberry Pi.

3. Development of an IoT-Based Smart Waste System with RTC Time Logging for Cleaning Assistance.

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